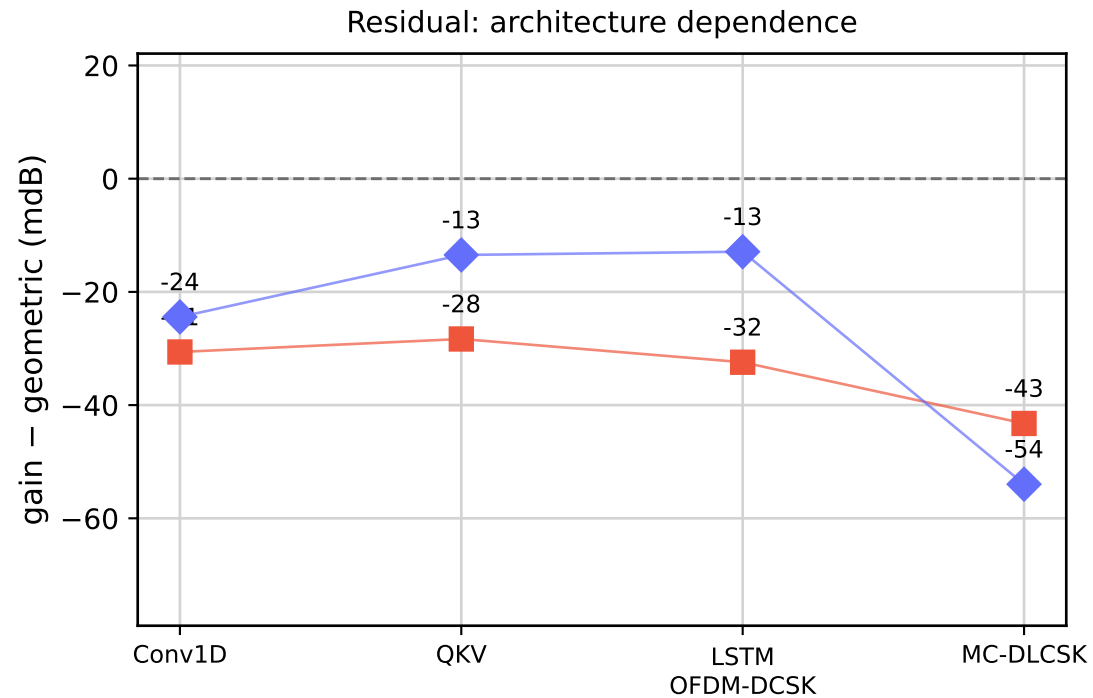
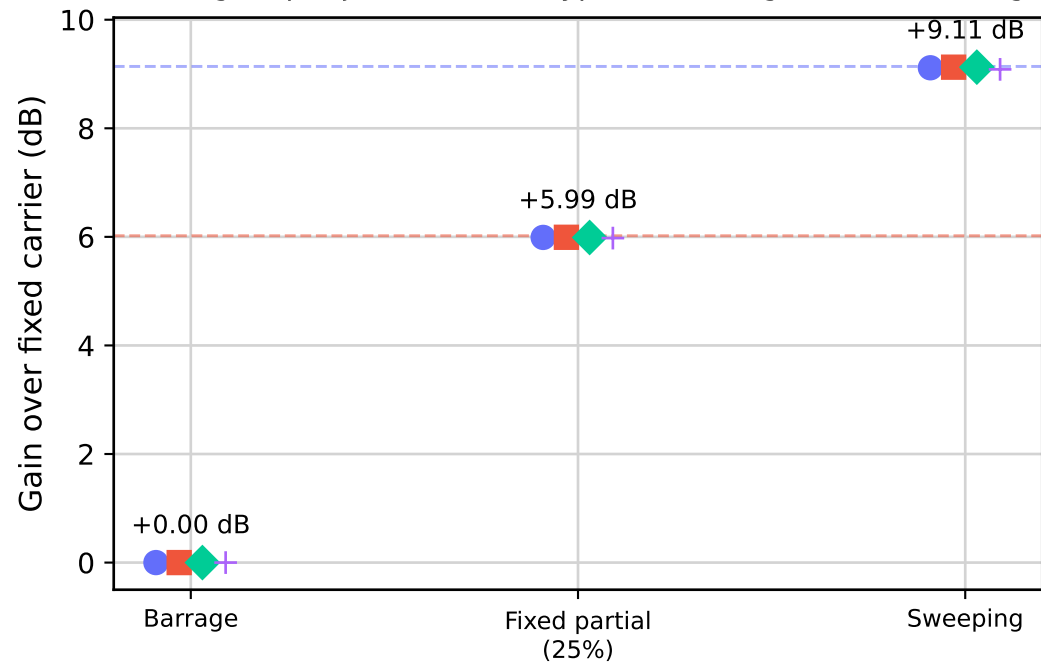


## Hopping gain over the fixed carrier (JSR = +10 dB)

Absolute gain per jammer archetype (dashed: geometric  $-10\log_{10} f$ )



- Ultra-CAN (Conv1D)
- Ultra-CAN (QKV)
- LSTM-OFDM-DCSK
- MC-DLCSK

- Fixed partial (25%)
- Sweeping